

Multivariate Analysis – Honours

Section 1 Exercises Solutions

Chapter 5

Homework exercise 5.1

1. Large sample 95% Simultaneous T^2 confidence intervals are given by

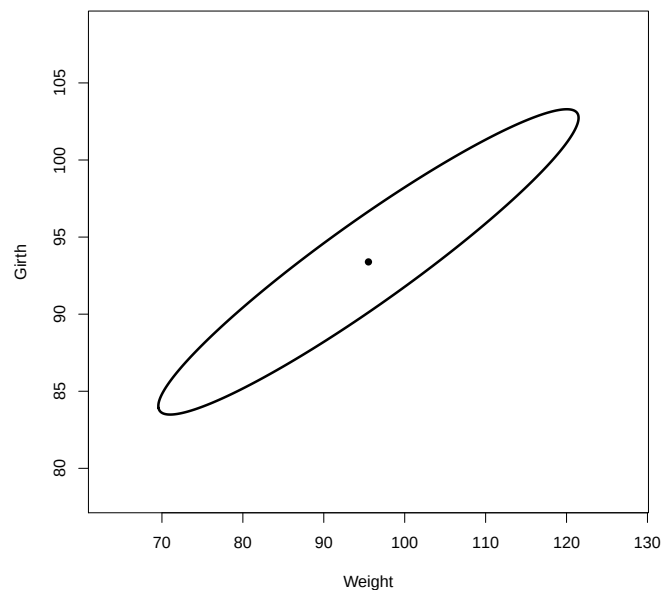
$$\bar{x}_i \pm \sqrt{\chi_6^2(0.05)} \sqrt{\frac{s_{ii}}{61}} = \bar{x}_i \pm \sqrt{12.59} \sqrt{\frac{s_{ii}}{61}},$$

yielding

	Lower	Upper
Weight	69.55	121.49
Body Length	152.17	176.59
Neck	49.61	61.77
Girth	83.49	103.29
Head Length	16.55	19.41
Head Width	29.04	33.22

2. The large sample 95% simultaneous confidence ellipse for Weight and Girth is determined by all μ_1 and μ_4 such that

$$\begin{bmatrix} 95.52 - \mu_1 & 93.39 - \mu_4 \end{bmatrix} S^{-1} \begin{bmatrix} 95.52 - \mu_1 \\ 93.39 - \mu_4 \end{bmatrix} \leq \frac{12.59}{61} = 0.2064$$



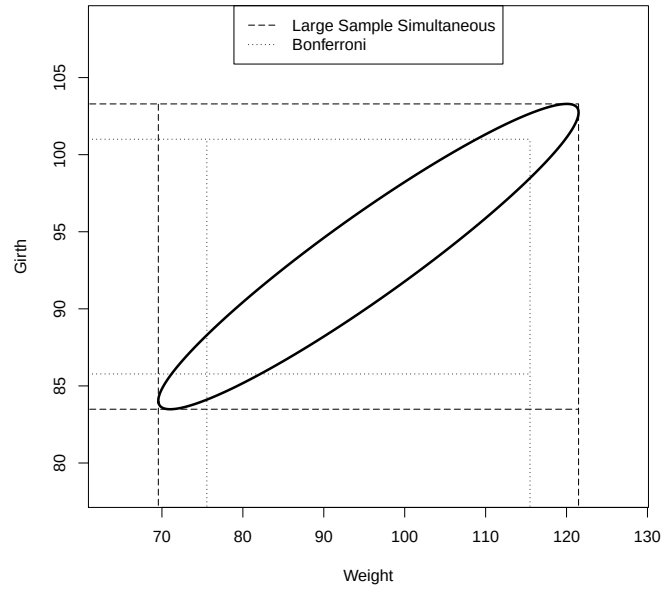
3. Bonferroni 95% simultaneous confidence intervals are given by

$$\bar{x}_i \pm t_{60}(0.025/6) \sqrt{\frac{s_{ii}}{61}} = \bar{x}_i \pm 2.729 \sqrt{\frac{s_{ii}}{61}},$$

yielding

	Lower	Upper
Weight	75.55	115.49
Body Length	154.99	173.77
Neck	51.01	60.37
Girth	85.78	101.00
Head Length	16.88	19.08
Head Width	29.52	32.74

4. Both the simultaneous confidence and Bonferroni rectangle are shown here:



Weight and Girth are highly correlated ($r = 0.9437$). Therefore, the 95% confidence ellipse is smaller and more informative than the 95% Bonferroni rectangle.

5. Adding an additional confidence statement on $\mu_6 - \mu_5$ and therefore using $m = 7$, yields the following 95% confidence interval:

$$\bar{x}_6 - \bar{x}_5 \pm t_{60}(0.025/7) \sqrt{\frac{s_{55} + s_{66} - 2s_{56}}{61}} = 13.15 \pm 2.785 \sqrt{\frac{9.95 + 21.26 - 2(13.88)}{61}}$$

or

$$12.49 < \mu_6 - \mu_5 < 13.81$$

Homework exercise 5.2

1. 95% Simultaneous T^2 confidence intervals for the lengths are given by

$$\bar{x}_i \pm \sqrt{\frac{(7-1)4}{7-4} F_{4,7-4}(0.05)} \sqrt{\frac{s_{ii}}{7}} = \bar{x}_i \pm \sqrt{72.94} \sqrt{\frac{s_{ii}}{7}},$$

yielding

	Lower	Upper
Lngth2	130.69	155.89
Lngth3	127.02	191.55
Lngth4	160.31	185.98
Lngth5	155.37	198.91

2. To obtain the annual mean increases, $\mu_j - \mu_{j-1}, j = 3, 4, 5$, define the contrast matrix

$$\mathbf{C} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

such that $\mathbf{C}\boldsymbol{\mu} = \begin{bmatrix} \mu_3 - \mu_2 \\ \mu_4 - \mu_3 \\ \mu_5 - \mu_4 \end{bmatrix}$

If we let $\bar{\mathbf{X}}_{\Delta} = \mathbf{C}\bar{\mathbf{X}}$ and $\mathbf{S}_{\Delta} = \mathbf{CSC}'$, then the resulting 95% simultaneous T^2 confidence intervals for the annual increase in lengths are given by

$$\bar{x}_{\Delta i} \pm \sqrt{\frac{(7-1)3}{7-3} F_{3,7-3}(0.05)} \sqrt{\frac{s_{\Delta ii}}{7}} = \bar{x}_{\Delta i} \pm \sqrt{29.66} \sqrt{\frac{s_{\Delta ii}}{7}}$$

yielding

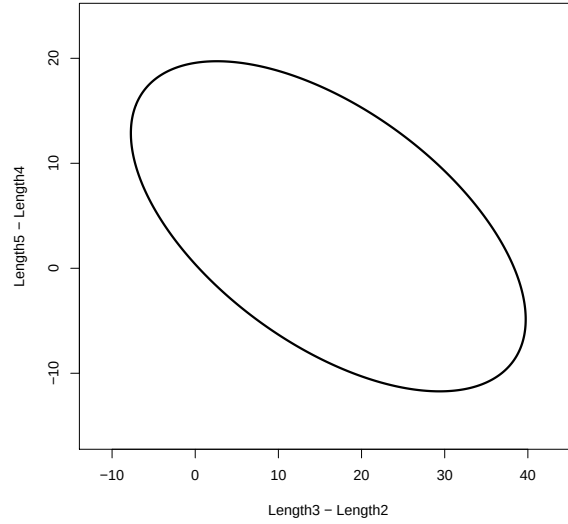
	Lower	Upper
Lngth3 - Lngth2	-7.74	39.74
Lngth4 - Lngth3	-9.48	37.19
Lngth5 - Lngth4	-11.72	19.72

3. The 95% simultaneous T^2 confidence ellipse for the mean increase in length from year 2 to 3 and the mean increase in length from year 4 to 5 is determined by all $\mu_3 - \mu_2$ and $\mu_5 - \mu_4$ such that

$$\begin{bmatrix} 16 - (\mu_3 - \mu_2) & 4 - (\mu_5 - \mu_4) \end{bmatrix} S_{\Delta}^{-1} \begin{bmatrix} 16 - (\mu_3 - \mu_2) \\ 4 - (\mu_5 - \mu_4) \end{bmatrix} \leq \frac{29.66}{7} = 4.237$$

where $S_{\Delta}^* = \{S_{\Delta jk}\}, j, k = 1, 3$

The ellipse is given below. Interestingly, we notice a strong negative correlation between the increase from year 2 to year 3 and the increase from year 4 to year 5 ($r = -0.83$) and a moderate negative correlation between the increase from year 3 to year 4 and the increase from year 4 to year 5 ($r = -0.56$). Therefore, the more these female grizzly bears grew in their 3rd year, the less they grew in subsequent years.



4. Let $\mathbf{Y}$ denote the set containing the four lengths and the three successive annual increases in length. The 95% Bonferroni confidence intervals for $\boldsymbol{\mu}_Y$ are given by

$$\bar{y}_i \pm t_6(0.025/7)\sqrt{\frac{s_{Yii}}{7}} = \bar{y}_i \pm 3.997\sqrt{\frac{s_{Yii}}{7}},$$

yielding

	Lower	Upper
Lngth2	137.39	149.18
Lngth3	144.19	174.39
Lngth4	167.14	179.15
Lngth5	166.95	187.33
Lngth3 - Lngth2	-1.42	33.42
Lngth4 - Lngth3	-3.27	30.98
Lngth5 - Lngth4	-7.54	15.54

5. We observe that the 95% Bonferroni rectangle is slightly smaller and more informative than the 95% confidence ellipse.

